

Lab 01

Registration No: 2025/CS/079

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01 .

Requirement	Description	Stakeholders	Why it is a High-Level	Challenge
Online payment gateway	Pay for meals using credit or debit cards	Customers	Focus on customer's privacy	Maintain a secure payment gateway option
Cash handling	Pay for meals using COD	Customers	Focus on customer's trust	Handling cash by a third party
Cart management	Stores user's choices in a. temporary storage	Customers	Focus on systems capability	Maintain a to uniquely store food items for users
Real-time order tracking	Customer tracking the delivery order	Customers	Focus on system's reliability	Delivering order in a given time period
Route management	Deciding the delivery route	Customers, Delivery Personals	Focus on efficiency	Choosing the best possible route

02 .

With abstraction we can ignore the complex requirements surrounds the system, focus on building the basic boiler plate of the system which handles all the basic tasks that are supposed fulfill by the system. After the completion of the initial phase we can keep building on top of the current system to add other required features as extensions for the current system.

03 .

Module Name	Primary Responsibility	Data Handled by the Module	Other Modules which interact
User Authentication	Identify users that are already on the system and creating new users.	Users' login info, Users' personal information	Cart generation, Order tracking
Cart generation	Create a cart for a user and stores all the items they place on it.	Meals user placed	Payment gateway, User Authentication, Restaurant browsing
Payment gateway	Create a pathway for user to make an online payment.	Total price of the order, User's card details	Cart generation, Order processing
Restaurant browsing	Create menus and individual food items for user to pick.	Food item, quantity	Cart generation
Order processing	After the payment is done, process the order and inform related parties	Order details from cart	Cart generation, Order tracking, Payment gateway
Order tracking	Tracking the order until it reaches the user	Users' location, Order details, Payment option, Tracking number	Payment gateway User authentication

04 .

- I) If the system is very large, it will surely have many complex requirements, so breaking down it into pieces will be easier to accomplish the task.
- II) Members can be assigned to work in different modules. This method will support teamwork since it enables the members to interact with each other to make the complete system, since the modules need to interact with each other.
- III) Testing the interaction between different processes will be harder if the system is developed as one large component. But if it was developed in modules the testing can easily be done by combining modules.
- IV) User authentication. If the user identification process is incomplete it might occur problems throughout the system.
- V) Restaurant browsing. Since it contains all the menu items, food categories, and individual food items, it will experience the highest system load.

05 .